

Brief A: Technical Maintenance Engineering

Objective: detect near-term pump failure from vibration, temperature, pressure, acoustic, current, and event-count data.

Metric	Result	Interpretation
Model	Bagged decision trees	Robust baseline for non-linear fault patterns.
Accuracy	94.7%	Share with sensitivity and false-alarm rate, not alone.
Sensitivity	81.9%	Percent of near failures caught before intervention window closes.
False alarm rate	1.0%	Healthy observations incorrectly escalated for maintenance review.
Shift t-test p-value	0.00613	Night and day vibration means differ at alpha = 0.05.
RUL MAE	109.0 hours	Average error for baseline remaining-useful-life prediction.

Recommended engineering action: deploy warning thresholds using vibration RMS, temperature trend, and pressure drop logic. Track confusion matrix movement weekly and tune the alert threshold if false alarms exceed operating tolerance.

Brief B: CFO / Executive Summary

Business issue: unplanned pump failures interrupt throughput, create overtime maintenance work, and increase safety exposure. The predictive maintenance model identifies assets likely to fail within seven days so work can be scheduled before downtime occurs.

Executive question	Answer
What changes operationally?	Maintenance shifts from calendar-based work to risk-based scheduling.
Why fund it?	The model targets avoided downtime, fewer emergency callouts, and better spare-parts planning.
Main risk control	It catches about 82% of near failures in the synthetic pilot.
Cost discipline	Avoid replacing equipment too early by using condition evidence rather than fixed age alone.
Success metric	Reduce avoidable downtime and false alarms while preserving production capacity.

Recommended executive action: approve a 90-day pilot on high-criticality pumps, measure avoided downtime, emergency maintenance hours, false alarms, and operator adoption, then decide whether to scale.